Sivakorn Khundilokrattaya

3rd-Year Computer Engineering Student • Seeking Data Science, ML & Quantitative Fintech Internship

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EDUCATION & SCHOLARSHIP

Srinakharinwirot University (SWU) – B.Eng. in Computer Engineering 2024 – Present (Expected 2028)

- **Focus & Specialization:** Statistical Machine Learning, Quantitative Credit Risk Modeling, Explainable AI (XAI), and Data Engineering.
- **Core Coursework:** Machine Learning, Probability & Statistics, Database Systems, Data Structures & Algorithms, Operating Systems, Software Engineering.

NHK Spring (Thailand) Undergraduate Scholarship – Corporate Engineering Scholar 2026 – 2027

- **Scholarship & Internship:** Awarded an undergraduate engineering scholarship (conferred Sep 2026 under cumulative GPAX ≥ 2.75 and faculty endorsement criteria), including a planned 2-month summer industrial internship at NHK Spring (Thailand) Co., Ltd.

TECHNICAL SKILLS & COMPETENCIES

ML & Quantitative Risk:	LightGBM, CatBoost, XGBoost, Scikit-Learn, PyG (R-GCN), Optuna, OptBinning (WoE/IV)
Explainable AI & Risk:	TreeSHAP (Reason Codes), DiCE Counterfactuals, PSI Drift Monitoring, KS & Brier Score
Programming & Data:	Python (Pandas, NumPy, SciPy), SQL (PostgreSQL, SQLite), TypeScript, JS, C/C++, Bash
Engineering & Serving:	FastAPI, ONNX, Docker, WebSockets, SQLAlchemy, React, Next.js, GeoPandas, Rasterio, Git
Languages:	Thai (Native), English (CEFR B1 – Oxford Placement Test), Japanese (Elementary)

FEATURED DATA SCIENCE & ML PROJECTS

K-NEXUS – FlowSense & TrustGraph (Synthetic Cashflow & Risk Prototype) • App | Showcase 2026

PyG (R-GCN), LightGBM, ONNX, FastAPI, React • KBTG Hackathon 2026 – Top 30 Overall & Top 10 Data Science

- **Leakage-Free Graph Ablation (TrustGraph):** Built a chronological past-only transaction graph so 16D R-GCN embeddings could not leak future mule labels. Evaluated on 1,354 synthetic holdout transfers (67 positives), showing that the tabular baseline (0.996 PR-AUC) outperformed R-GCN + LightGBM (0.994 PR-AUC).
- **Validation Gating & Model Rejection (FlowSense):** Evaluated 1, 7, 14, and 30-day cashflow candidates via paired cumulative MAE on 32 synthetic accounts (30-day candidate MAE reached 2,701.6 THB vs. 313.3 THB account-mean baseline across 1,121 origins) and kept FlowSense ML disabled after failing Gate 2 criteria.
- **Runtime Architecture & Stage Pitch:** Built and pitched the FastAPI + React 18 prototype, serving an explainable history-mean, scenario-fallback, or abstention rule alongside offline ONNX Runtime transfer risk review.

Credit Risk Scoring & Explainable AI (XAI) Decision-Support Prototype • GitHub Repo 2026

Python, LightGBM, Optuna, OptBinning (WoE/IV), TreeSHAP, DiCE, FastAPI, Next.js • Research Prototype

- **Monotonic Feature Screening:** Built a Probability of Default (PD) pipeline on Home Credit and bureau datasets using domain financial ratios, applying monotonic Weight of Evidence (WoE) binning via OptBinning to enforce economic directionality while screening for Information Value (IV ≥ 0.02) and multicollinearity (VIF < 5.0).
- **Baseline Benchmarking & Scorecard Calibration:** Tuned a 5-fold stratified LightGBM ensemble with Optuna against an interpretable WoE logistic regression baseline, improving cross-validation ROC-AUC from 0.742 to 0.785 with 43.2% KS separation, and calibrated predictions to a 300–850 point scorecard (Base 600, PDO 20).
- **Explainability & Audit Workflow:** Integrated TreeSHAP adverse-action reason codes, DiCE counterfactual recourse scenarios, simulated Population Stability Index (PSI) drift checks, and SQLite audit logging into a FastAPI + Next.js underwriting dashboard.

FEATURED DATA SCIENCE & ML PROJECTS (CONTINUED)

Shopee Smart COD Reliability & Adaptive Checkout Prototype (Synthetic Data) • GitHub Repo 2026

Python, Scikit-Learn (Random Forest), FastAPI, JS • Shopee Ultimate Case Challenge 2026 – Operations Strategy

- **Statistical Formulation & Calibration:** Evaluated simulated COD vs. prepaid failure disparity via two-proportion Z-test (10.6× ratio, $z = 24.4$, $p < 0.001$) across 50,000 synthetic orders. Trained a 3-fold sigmoid-calibrated Random Forest achieving 0.840 ROC-AUC, 0.306 PR-AUC, and 0.021 Brier score on 3,500 held-out synthetic orders.
- **Adaptive Checkout Policy & Feedback Loop:** Mapped predicted failure probabilities into bounded reliability-score adjustments (+1 / -3 / -6 pts within 5–99) across a 4-tier checkout policy (Grades A–D: 1-click COD, reminder, 20% deposit, or 30% deposit/prepaid), paired with delivery-feedback updates (+8/-20 pts) and rule-based seller checks.
- **Scenario Modeling & Simulator:** Built a FastAPI scoring service and mobile checkout simulator modeling 3.58M THB/month in simulated return-cost savings (assuming a target 58% RTO failure reduction).

Long Overdue Debtor (LOD) Default Prediction Pipeline • GitHub Repo 2025

CatBoost, LightGBM, XGBoost, Optuna, Scikit-Learn • Aihack Thailand 2025 – National Finalist (Team OEIA)

- **Imbalanced Financial Feature Engineering:** Engineered leak-free financial ratios (debt-to-income burden, credit utilization, unused credit headroom, tenure stability, and product debt density) and explicit missingness indicators for 12-month default prediction on confidential consumer loan records (~8.3% default rate / 1:12 imbalance).
- **Out-of-Fold Validation & Blending:** Implemented cohort-aware 5-fold stratified cross-validation and out-of-fold probability/rank blending across CatBoost and LightGBM, improving OOF ROC-AUC from 0.633 baseline to 0.651 (transforms) and 0.653 (final blend).
- **Finalist Pitch & Reproducible Pipeline:** Presented risk-cutoff trade-offs in the National Finalist round and published an NDA-safe synthetic data pipeline so the preprocessing and blending workflow can be inspected without exposing confidential lending records.

Industrial IoT Telemetry & OEE Data Platform (Simulated Telemetry) • GitHub Repo 2026

Simulated sensor-telemetry pipeline (Python, FastAPI, PostgreSQL 15, WebSockets, React 18) computing hourly OEE summaries (Availability × Performance × Quality) and Isolation Forest / LightGBM / TreeSHAP anomaly views.

OTHER HACKATHONS & APPLIED AI CHALLENGES

GISTDA Hackathon (Spatial Data Science Developer): Sentinel-1 SAR & DEM flood-hazard pipeline. Jan 2026

True Innovation Launchpad 2026 (AI & Full-Stack Developer): GuardianAI frailty triage • GitHub Jan 2026

STANDARDIZED ASSESSMENTS & CREDENTIALS

SKILLKAMP by KBTG – Data Analyst Assessment (STAMP) Mar 2026

Verified score of 97/150 (Above KBTG Industry Benchmark of 90).

Oxford Placement Test – English Proficiency (AUA Language Center / Oxford University Press) Jun 2026

Verified CEFR B1 (Independent User, Overall Score: 51/120 • Use of English: 54 / B1, Listening: 48 / B1).